Baseline Neural Network Study for Learning Constitutive Laws in Granular Flow

Name : Abhishek Tagalpallewar || 2nd Year Undergraduate || Course code : ME 299 || Institute : IIT Gandhinagar || Supervisor : Harkirat Singh

Poster ID : 6

Introduction

Problem Statement

To establish a baseline neural network model, we trained neural networks to approximate simple nonlinear functions such as quadratic and sine functions. This helps evaluate how well different network architectures learn nonlinear relationships before applying them to complex simulation data.

Neural Network

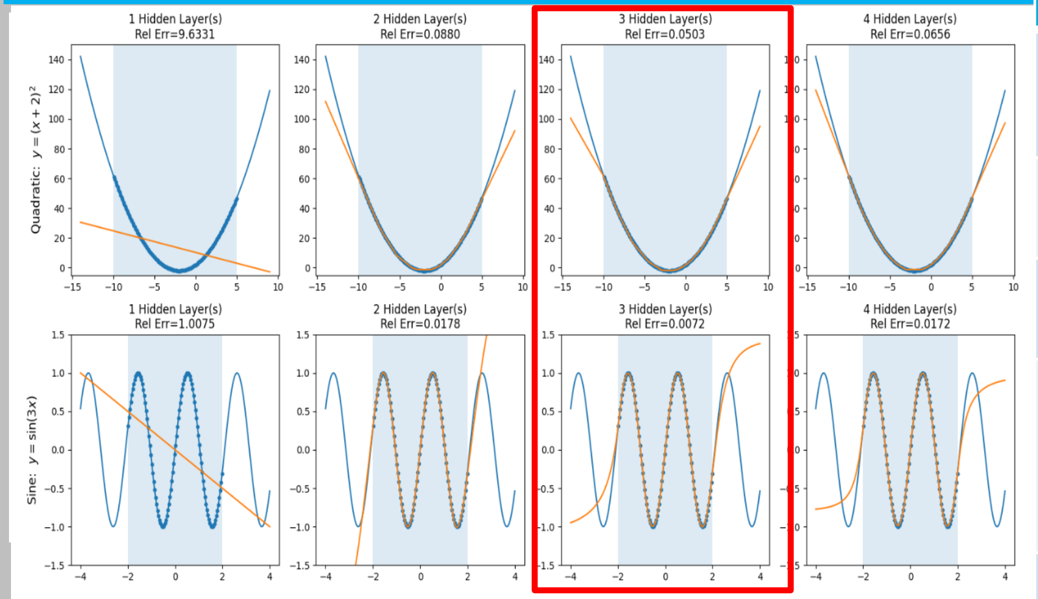
A neural network is a computational model that learns patterns in data by adjusting weights and biases applied to inputs through interconnected layers of neurons. During training, the predicted output is compared with the true value using a loss function, and the weights and biases are updated through backpropagation to minimize the prediction error.

$$Loss = \frac{1}{N}\sum(True\ Value - \overline{Predicted\ Value})^2$$

Future Work

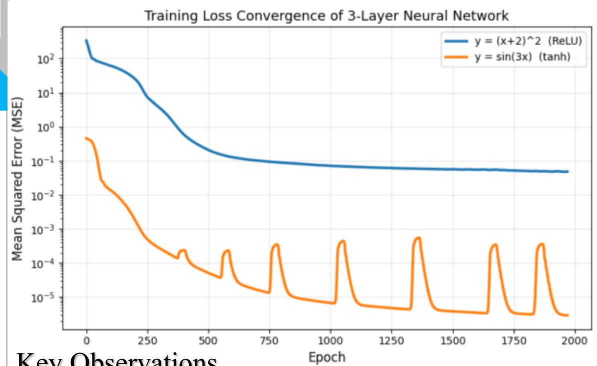
- Explore additional activation functions and network architectures to improve model accuracy and learning capability.
- Evaluate different error metrics such as Mean Squared Error (MSE) and Relative Error to better assess model performance.
- Train the neural network model using the Discrete Element Method (DEM) simulation data (Poster ID: 86) to learn constitutive laws for granular flow.
- Test the neural-network-based constitutive model on a new DEM data.

Plots



Aspect	Explanation
Selected Depth	A 3-Layer Neural Network provides sufficient capacity to model nonlinear functions while avoiding unnecessary model complexity.
Effect of Shallow Networks	1–2 layer networks underfit and fail to capture curvature and oscillatory behavior effectively.
Effect of Deeper Networks	4 layers provide minimal improvement while increasing model complexity.
Activation Function Choice	• ReLU is used for the quadratic function as it efficiently approximates polynomial curvature through piecewise linear behavior. • tanh is used for the sine function because its smooth nonlinear shape is well suited for modeling periodic patterns
Overall Observation	The selected architecture achieves low training error and accurate interpolation within the training region.

Results



Key Observations

- The neural network accurately learns both functions within the training (interpolation) region.
- The training MSE decreases with epochs, indicating successful learning through backpropagation.
- Predictions in the extrapolation region are less accurate, showing that neural networks generally perform better at interpolation than extrapolation

Parameter	Quadratic Function (y = x^2 + 4x + 4)	Sine Function (y = sin(3x))
Network Architecture	3-Layer Neural Network	3-Layer Neural Network
Neurons per Hidden Layer	20	20
Activation Function	ReLU	tanh
Loss Function	Mean Squared Error (MSE)	Mean Squared Error (MSE)
Backpropagation	Gradient-based weight and bias updates	Gradient-based weight and bias updates
Training Domain	(x in [-10,5])	(x in [-2,2])
Extrapolation Region	(x in [-14,-10] and [5,9])	(x in [-4,-2] and [2,4])
Training Epochs	2000	2000
Relative Error	0.0503	0.0072